

1. Administrative & Study Identification

Full Study Title: A Study of Dato-DXd Versus Investigator's Choice Chemotherapy in Patients With Locally Recurrent Inoperable or Metastatic Triple-negative Breast Cancer, Who Are Not Candidates for PD-1/PD-L1 Inhibitor Therapy (TROPION-Breast02)
Official Title: A Phase 3, Open-label, Randomised Study of Datopotamab Deruxtecan (Dato-DXd) Versus Investigator's Choice of Chemotherapy in Patients Who Are Not Candidates for PD-1/PD-L1 Inhibitor Therapy in First-line Locally Recurrent Inoperable or Metastatic Triple-negative Breast Cancer (TROPION Breast02)
Short Title/Acronym: TROPION-Breast02 **Protocol Number:** D926PC00001 **NCT Number:** NCT05374512 **Sponsor/Company:** AstraZeneca **Investigational Product:** Datopotamab deruxtecan (Dato-DXd) **Phase of Development:** Phase III **Trial Status:** Active, not recruiting **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To demonstrate the superiority of Dato-DXd relative to Investigator's Choice Chemotherapy (ICC) by assessment of Progression-Free Survival (PFS) per Blinded Independent Central Review (BICR) and Overall Survival (OS). **Secondary Objectives:** To evaluate the Objective Response Rate (ORR), Duration of Response (DoR), Progression-Free Survival (PFS) by Investigator assessment, health-related quality of life (HRQoL), safety, and tolerability.

2.2 Background & Rationale To address the critical unmet medical need in patients with locally recurrent inoperable or metastatic Triple-Negative Breast Cancer (TNBC) for whom immunotherapy (PD-1/PD-L1 inhibitors) is not an option. Despite recent advances, prognosis remains poor for advanced TNBC, and this study aims to evaluate Dato-DXd—a TROP2-directed antibody-drug conjugate (ADC)—as a more effective and tolerable first-line standard of care compared to traditional chemotherapy.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Investigator's Choice Chemotherapy: paclitaxel, nab-paclitaxel, carboplatin [Trade Name: Paraplatin, ATC Code: L01XA02], capecitabine, or eribulin) **Blinding:** Open-label **Randomization:** 1:1

3.2 Planned Enrollment & Duration Targeted Enrollment (\$N\$): 644 patients **Number of Sites:** Global and Regional breakdown (Multinational: USA, Europe, Asia, South America, etc.) **Estimated Study Start:** 16-May-2022 **Estimated Primary Completion:** 25-Aug-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years at the time of screening. 2. Histologically/cytologically confirmed locally recurrent inoperable or metastatic triple-negative breast cancer (TNBC), defined as negative for ER, PR, and HER2. 3. Previously untreated with chemotherapy or other systemic anti-cancer therapy for metastatic or locally recurrent inoperable disease, and not a candidate for PD-1/PD-L1 inhibitor therapy. 4. ECOG Performance Status 0 or 1 with no deterioration over the previous 2 weeks. 5. Minimum life expectancy of 12 weeks.

4.2 Exclusion Criteria 1. History of another primary malignancy (except for malignancy treated with curative intent with no known active disease within 3 years). 2. Current or prior non-infectious interstitial lung disease (ILD)/pneumonitis that required steroids, or clinically severe pulmonary function compromise. 3. Prior treatment with topoisomerase I inhibitors (including ADCs), TROP2-targeted therapy, or the same chemotherapy agent chosen for on-study ICC. 4. Known active or uncontrolled hepatitis B or C virus infection, or known human immunodeficiency virus (HIV) infection that is not well controlled.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Dato-DXd 6 mg/kg administered every 3 weeks. **Control Regimen:** Investigator's Choice Chemotherapy (paclitaxel, nab-paclitaxel, carboplatin, capecitabine, or eribulin) based on investigator assessment. **Route of Administration:** Intravenous (IV) or Oral (depending on the agent, e.g., capecitabine). **Duration of Treatment:** Until disease progression, unacceptable toxicity, or patient withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care medications for the management of therapy-related adverse events (e.g., antiemetics). **Prohibited:** Concurrent anti-cancer therapies, systemic hormone replacement therapy, and live attenuated vaccines within 30 days prior to the first dose.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline Tumor Imaging, Labs
Baseline	Day 0	Randomization, First Dose Administration
Interim	Every 3 Weeks (Q3W)	Treatment cycles, Safety Labs, Efficacy Assessment (Tumor Imaging), HRQoL
End of Study	Upon progression/toxicity	Final Vital Signs, Exit Interview, IP Accountability, Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression-Free Survival (PFS) and Overall Survival (OS). **Measurement Method:** PFS assessed by Blinded Independent Central Review (BICR) per RECIST version 1.1; OS measured as the time from randomization to death from any cause.

7.2 Secondary & Exploratory Endpoints - Confirmed Objective Response Rate (ORR) - Duration of Response (DoR) - Progression-Free Survival (PFS) by Investigator assessment

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous evaluation via patient report, clinical examination, and laboratory assessments (monitoring closely for stomatitis, nausea, fatigue, neutropenia, and ILD/pneumonitis). **Serious Adverse Events (SAEs):** Reported within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Independent external committee conducting periodic safety and efficacy reviews.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) Set for efficacy analyses (all randomized patients); Safety Set for patients receiving at least one dose of the study intervention. **Sample Size Justification:** Appropriately powered to detect statistically significant superiority in both PFS and OS (α split between endpoints). **Handling of Missing Data:** Censoring rules for PFS per FDA/EMA oncology guidelines; standard imputation for missing HRQoL data.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring visits by the sponsor/CRO. **Audit Trail:** Validated Electronic Data Capture (EDC) system with full audit trails and electronic signature capabilities.

11. Study Identifiers & Governance

Primary ID: D926PC00001 **Registry Number:** NCT05374512 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** TROPION-Breast02 **Official Regulatory Title:** A Phase 3, Open-label, Randomised Study of Datopotamab Deruxtecan (Dato-DXd) Versus Investigator's Choice of Chemotherapy in Patients Who Are Not Candidates for PD-1/PD-L1 Inhibitor Therapy in First-line Locally Recurrent Inoperable or Metastatic Triple-negative Breast Cancer (TROPION Breast02)

12. Clinical Framework (TNBC Specific)

Disease Area: Breast Cancer **Preferred Disease Terminology:** Malignant neoplasm of breast / Triple Negative Breast Neoplasms **Primary Condition:** Triple-Negative Breast Cancer (TNBC) **Diagnosis Threshold:** Histologically/cytologically confirmed ER-, PR-, and HER2-negative breast cancer **Medical Methodology:** Targeted TROP2-directed Antibody-Drug Conjugate (ADC) vs. Chemotherapy **Optimization Target:** PFS and OS improvement in a first-line metastatic setting

13. Study Procedures & Methodology

Management Pathway: Standard oncology clinical pathway for advanced TNBC **Education Protocol (HCP):** Site initiation visits covering protocol compliance, RECIST 1.1 tumor evaluation, and ILD/pneumonitis risk management **Education Protocol (Patient):** Informed consent process detailing the IV administration schedule, potential side effects, and required follow-up **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) for efficacy endpoints

14. Observation & Follow-Up Schedule

Total Duration: Follow-up until study completion (Estimated Nov-2028) **Onsite Visit Cadence:** Every 3 weeks (Q3W) correlating with the treatment cycles **Remote Monitoring Frequency:** As per sponsor clinical monitoring plan **Checklist Review Interval:** Tumor assessments conducted every 6 to 9 weeks per RECIST 1.1

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]					